

Multiple Choice

1. **Answer:** C
Options: A) 2; B) 4; C) 8; D) 16
Note: Correct option: C - 8
2. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
3. **Answer:** B
Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.
Note: Correct option: B - Improve the optimisation algorithm being used for error minimisation.
4. **Answer:** A
Options: A) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$; B) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$; C) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$; D) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$
Note: Correct option: A - $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$
5. **Answer:** A
Options: A) $p(y|x, w)$; B) $p(y, x)$; C) $p(w|x, w)$; D) None of the above
Note: Correct option: A - $p(y|x, w)$

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'underfitting'.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Whether we assume full class covariance matrices or diagonal class covariance matrices. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** `5 + torch.randn(10,5) * 4; 2 * torch.rand(10,10) - 1`. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key